Artificial Intelligence Course Syllabus IN ACTE

Module 1: Introduction to Data Science

- What is Data Science?
- What is Machine Learning?
- What is Deep Learning?
- What is AI?
- Data Analytics & it's types

Module 2: Introduction to Python

- What is Python?
- Why Python?
- Installing Python
- Python IDEs
- Jupyter Notebook Overview

Module 3: Python Basics

- Python Basic Data types
- Lists
- Slicing
- IF statements
- Loops
- Dictionaries
- Tuples
- Functions
- Array
- Selection by position & Labels

Module 4: Python Packages

- Pandas
- Numpy
- Sci-kit Learn

- Mat-plot library

Module 5: Importing Data

- Reading CSV files
- Saving in Python data
- Loading Python data objects
- Writing data to csv file

Module 6: Manipulating Data

- Selecting rows/observations
- Rounding Number
- Selecting columns/fields
- Merging data
- Data aggregation
- Data munging techniques

Module 7: Statistics Basics

- Central Tendency
- Probability Basics
- Standard Deviation
- Bias variance Trade off
- Distance metrics
- Outlier analysis
- Missing Value treatment
- Correlation

Module 8: Error Metrics

- Classification
- Regression

Module 9: Machine Learning

- Supervised Learning
- Linear Regression
- Logistic regression

Module 10: Unsupervised Learning

- K-Means
- K-Means ++
- Hierarchical Clustering

Module 11: SVM

- Support Vectors
- Hyperplanes
- 2-D Case
- Linear Hyperplane

Module 12: SVM Kernal

- Linear
- Radial
- polynomial

Module 13: Other Machine Learning algorithms

- K – Nearest Neighbour
- Naïve Bayes Classifier
- Decision Tree – CART
- Decision Tree – C50
- Random Forest

Module 14: ARTIFICIAL INTELLIGENCE

- Perceptron
- Multi-Layer perceptron
- Markov Decision Process
- Logical Agent & First Order Logic
- AL Applications

Module 15: Deep Learning Algorithms

- CNN – Convolutional Neural Network
- RNN – Recurrent Neural Network

- ANN – Artificial Neural Network

Module 16: Introduction to NLP

- Text Pre-processing
- Noise Removal
- Lexicon Normalization
- Lemmatization
- Stemming
- Object Standardization

Module 17: Text to Features

- Syntactical Parsing
- Dependency Grammar
- Part of Speech Tagging
- Entity Parsing
- Named Entity Recognition
- Topic Modelling
- N-Grams
- TF – IDF
- Frequency / Density Features
- Word Embedding's

Module 18: Tasks of NLP

- Text Classification
- Text Matching
- Levenshtein Distance
- Phonetic Matching
- Flexible String Matching
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